



National Institute of Technology, Kurukshetra
Department of Computer Engineering
Branch: CS, IT

End Semester Examination

Date of Exam – 9-August-2026

Time: 02:00 PM-5:00 PM [3 Hours]

Max Marks:50

Programme: **B.Tech (Computer Engineering)**

Subject: **Scripting Languages**

Roll Number: 124103030

Semester: **4th**

Subject Code: **CSPE214/ITPE214**

Notes:

- There are two pages in this question paper and all questions are compulsory.
- Unless stated otherwise, the symbols have their usual meanings in the context of the subject. Assume suitably and state additional data required, if any.
- Before starting to write the solutions, candidates should check the question paper for any discrepancy and ensure that they have been delivered the question paper of the right course code.
- Your code should be clearly written, well commented and have logical variable names.

Sr. No.	Question	Marks																	
Q.1. (a) (b)	A translator is a program that converts source code into machine code in scripting languages. In this regard, Contrast between Compilers and Interpreters. Write a Bash script that defines a function to reverse a number passed as a command-line argument using a while loop.	[3] [7]																	
Q.2. (a) (b)	How is HTML different from HTML5? Give an example by writing code. Develop an HTML webpage to display your coding progress in the format shown in Table 1 below.	[3] [7]																	
Table 1: Coding Tracker																			
<table><tr><th>Platform</th><th>Problems Solved</th><th>Difficulty</th><th>Streak</th></tr><tr><td>LeetCode</td><td>180</td><td>Medium</td><td>25 Days</td></tr><tr><td>HackerRank</td><td>95</td><td>Easy</td><td>15 Days</td></tr><tr><td>CodeChef</td><td>120</td><td>Medium</td><td>12 Days</td></tr></table>				Platform	Problems Solved	Difficulty	Streak	LeetCode	180	Medium	25 Days	HackerRank	95	Easy	15 Days	CodeChef	120	Medium	12 Days
Platform	Problems Solved	Difficulty	Streak																
LeetCode	180	Medium	25 Days																
HackerRank	95	Easy	15 Days																
CodeChef	120	Medium	12 Days																
Q.3. (a) (b)	Describe the three methods of applying CSS to an HTML document. Illustrate each method with an example and explain the order of precedence (priority) among them. Design a weather forecast card as shown below in Figure 1 displaying the city name, weather icon, temperature, humidity, wind speed, and condition using HTML and CSS only.	[3] [7]																	
Kurukshetra  32°C <table><tr><th>Parameter</th><th>Value</th></tr><tr><td>Humidity</td><td>70%</td></tr><tr><td>Wind Speed</td><td>12 km/h</td></tr><tr><td>Condition</td><td>Sunny</td></tr></table>			Parameter	Value	Humidity	70%	Wind Speed	12 km/h	Condition	Sunny									
Parameter	Value																		
Humidity	70%																		
Wind Speed	12 km/h																		
Condition	Sunny																		
Figure 1: Weather Forecast Card																			

Figure 1: Weather Forecast Card

Q.4. (a) (b)	What would be the output of the following code snippet? <pre> var x = 5; 8 7 var y = 8; 7 6 var z = x++ + --y + ++x + y--; console.log(x); 7 console.log(y); 6 console.log(z); 26 </pre> Write a JavaScript function to compute the value of b^n where n is the exponent and b is the base. Accept b and n from the user and display the results.	[3] [7]
Q.5. (a) (b)	Differentiate between sessions and cookies in PHP. Explain their key differences and illustrate your answer with the appropriate PHP syntax for starting a session and creating a cookie. Develop a PHP program that checks whether an email address belongs to the NIT Kurukshetra domain. Use the strpos() function only to verify whether the email address contains "@nitkkr.ac.in" and display an appropriate message. Sample Input: student@nitkkr.ac.in Sample Output: Valid NITKKR Email Address Alternative Output (if the email does not belong to the domain): Invalid NITKKR Email Address	[3] [7]

-----END OF PAPER-----